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Final Report

Origins of Mars's Moons: Physics-Structured ML Using Asteroid Tidal-Disruption SPH Simulations

by

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Table of Contents

List of Tables

List of Figures

Abstract

One theory for the formation of Mars's moons, Phobos and Deimos, is that close collisions between asteroids and Mars can tidally damage the asteroid and leave some debris bonded to the planet. This process is thoroughly modelled by smoothed-particle hydrodynamics (SPH), but testing every possible combination of inputs is computationally costly. In this study, bound mass fraction (BMF) is predicted from the simulation inputs using machine learning models after SPH outputs are converted into fragmentation measurements. The primary model, a tuned Gradient Boosting regressor using the original simulation inputs, achieves a grouped cross-validated R^2 of 0.9234, MAE of 0.0159, and RMSE of 0.0257. Adding physics-derived features gives no meaningful extra improvement, so the simpler tuned raw-input model is retained. The models are assessed using grouped cross-validation. In a controlled set, predicted BMF generally decreases from roughly 0.27 at 1.2 Mars radii to approximately zero by about 2.5 Mars radii as periapsis increases. A similarly clear decline occurs as encounter speed increases, from roughly 0.25 at low velocity to approximately zero by about 1.2 km/s. While the impacts of spin and velocity depend on the encounter conditions, periapsis is the strongest overall physical input. As a result, distinct physical inputs may become more significant in different regions of the measured parameter range. Changes in spin in matched cases result in BMF discrepancies of up to 0.24 at bigger periapsis and between 0.06 and 0.09 below 1.5 Mars radii. Despite falling within the numerical training ranges, a sparsely sampled 1019.5 kg instance yields held-out errors close to 0.08 BMF, demonstrating that being within the input range does not ensure a trustworthy prediction. Therefore, the model does not replace SPH or prove physical causality, but it is helpful for quickly screening well-supported encounter circumstances and highlighting parameter correlations that call for additional SPH research.

1. Introduction

1.1 Research Problem and Questions

The origin of Phobos and Deimos, the moons of Mars, is still a mystery. A close asteroid contact with Mars is one suggested route. The asteroid may be stretched or broken apart by variations in Martian gravity. While some debris might escape, others might stay gravitationally attached to Mars. Later orbital evolution and accretion into moons may benefit from this retained material (Kegerreis et al., 2024).

Smoothed-particle hydrodynamics (SPH) works well in this case because its particles move with the material as it deforms, fragments, and interacts gravitationally. However, simulating each new arrangement is computationally expensive. It is simply not practical to test every possible combination of asteroid mass, closest approach, encounter speed, spin, and numerical parameters. Using the present simulation archive, this study creates a machine learning (ML) surrogate to filter probable scenarios and identify whether new SPH simulations are necessary.

There are three research questions addressed here: (1) how do orbital and physical parameters affect disruption and bound retention; (2) which inputs matter most for retained debris; and (3) can machine-learning models predict outcomes within the sampled data while showing when a prediction should or should not be trusted?

1.2 Significance and Originality

The work has both scientific and computational significance. From a scientific perspective, simulations of a mechanism that might provide debris to the Martian system can be found in the SPH archive. It solves a frequent computational issue in simulation-based science: while a thorough solver can accurately simulate one setup, it is computationally costly to run multiple setups. Full SPH simulations can be concentrated on situations where ML models are most helpful because, once trained, they can swiftly analyse many combinations.

The primary contribution is the integration of physical interpretation with unambiguous testing for the reliability of an ML prediction. Each simulation's particle-level SPH outputs are combined into a single set of results by the workflow, which also tests the model on physical simulations that weren't used for training, compares matched and controlled cases, calculates the amount of similar SPH data that is present nearby, and uses these checks to suggest whether a new SPH run is necessary.

1.3 Literature Review and Research Gaps

The formation of Mars' moons, Phobos and Deimos, have been explained in several ways. Debris disks surrounding Mars can be created by giant-impact simulations (Citron et al., 2015; Canup & Salmon, 2018). Phobos and Deimos may originate from a disrupted shared parent body, according to orbital studies, and temporary inner moons may influence how material in the outer disk accretes (Rosenblatt et al., 2016; Bagheri et al., 2021). Nevertheless, little is known about the mechanisms that create and hold onto debris in the vicinity of Mars. So, by demonstrating that a close asteroid contact with Mars can tidally destroy the asteroid while trapping part of its debris, Kegerreis et al. (2024) offer a significant alternative creation pathway. The current project is directly motivated by this. Although their simulations demonstrate significant amounts of bound debris across various periapsis, velocity, mass, and spin settings, each new configuration necessitates a new simulation. This study investigates whether an ML model that screens new configurations prior to SPH execution can be trained using the current archive.

Since SPH is used to simulate these tidal-disruption scenarios, it is also crucial to comprehend how the simulation data are generated and handled. Because SPH is a mesh-free Lagrangian approach, computational particles follow the motion of the material they represent (Price, 2012). It is therefore appropriate for significant deformation and fragmentation. However, the subsequent processing of the simulation also affects the measured pieces. Here, post-processing techniques like friends-of-friends (FoF) use a selected connecting length to group neighbouring particles; therefore, the reported number and mass of fragments depend in part on the encounter physics as well as that processing option (More et al., 2011). Bound mass is handled differently: a fragment's orbital energy in relation to Mars determines whether it is bound or unbound.

The particle-level SPH outputs are transformed into scalar outcomes through post-processing, creating a compact tabular dataset appropriate for machine learning. Because tree-based models perform well with tiny, nonlinear tabular datasets, this makes the problem a good fit for machine learning (ML) (Grinsztajn et al., 2022). In the past, outcomes from comparable SPH collision and giant-impact simulations have also been predicted using tree-based ML surrogate models (Cambioni et al., 2019; Timpe et al., 2020). They are still not widely used for asteroid tidal-disruption encounters, though.

2. Methodology

2.1 Data Archive, Inputs, and Post-Processing

489 SPH outputs totalling over 218 GB of HDF5 particle data are included in the collection. These particle outputs are transformed into fragment and bound-mass measurements through post-processing. The asteroid and input parameter values are provided by the simulation setup and bound-mass findings are computed after post-processing breaks up the particle data. The main outcome/model target in this project is the bound mass fraction, represented by fbound, which is the fraction of the original asteroid mass that remains gravitationaly bound to Mars after the encounter, this will be referred to as BMF throughout this report.

Table 1. Main SPH setup and post-processing parameters.
Some parameter values appear much more often than others in the archive.

Parameter	**Coverage range**	**What it means**
Asteroid mass	A_{1800}-A_{2100}; A_{2000} dominates	Mass of the incoming asteroid. For example, $A_{2000} = 10^{20}$ kg. Its effect on BMF also depends on the other encounter conditions.
Periapsis	r_{11}-r_{30}; r_{12} and r_{16} common	Closest distance between the asteroid and Mars. For example, $r_{12} = 1.2\ R_{Mars}$ (Mars-radii). Smaller values mean a closer encounter and stronger tidal forces.
Encounter speed	v_{00}-v_{30}; v_{00} dominates	Speed of the asteroid during the encounter. v_{00} is the lowest-speed case, while higher v codes represent faster encounters. Higher speeds cause less capture.

Spin	none, x, y, z, mz; unevenly sampled	Rotation of the asteroid and the direction of its spin axis: none means no spin, x or y means equatorial spin, z means prograde z-spin, and mz means retrograde z-spin. Spin can change how the asteroid breaks up, but the different spin settings are unevenly sampled.
Resolution	n_{50}-n_{70}; n_{65} dominates	Number of SPH particles used in the simulation. For example, n_{65} is roughly $10^{6.5}$ particles. This is a numerical setting rather than a physical property of the encounter.
FoF linking length *(post-processing)*	0.0001-0.0126; 0.004 common	The maximum distance used to decide if nearby particles belong to the same fragment or not. Can change the measured fragments, does not change the physical encounter.

How the post-processing works is that FoF first groups nearby debris particles into fragments. The mass of each group g is equal to the sum of the particle masses, $Mg = \Sigma_i m_i$. The group's particular orbital energy with respect to Mars is then determined using its centre-of-mass position and velocity. A group is bound when $\varepsilon_g < 0$ and unbound when $\varepsilon_g \geq 0$. The number of bound fragments, the greatest and average bound-fragment mass, the total fragment count, and the largest-remnant measurements are among the exploratory outputs; BMF is the primary recorded output. It's also crucial to remember that BMF does not yet determine later disk or moon development, it merely measures gravitational retention.

2.2 Prediction Model Choices and Rationale

The main ML task is to predict the BMF value, or the proportion of the initial asteroid mass that is expected to stay bound to Mars, is the primary machine learning job. This maintains the distinction between retained fractions that are tiny, medium, and big. Furthermore, because Kegerreis et al. (2024) discovered that "tens of percent of an unbound asteroid's mass can be captured and survive beyond collisional timescales", a 10% BMF threshold is used as a straightforward screening criterion for higher-retention instances. Therefore, the 10% barrier is not a real physical limitation for moon formation, rather it is only utilized to classify and prioritize occurrences.

For the prediction tasks, several baseline models were used. While this project primarily focuses on tree-based models, Logistic Regression is also used for the BMF ≥ 10% classification task because it offers a straightforward baseline for testing whether lower- and higher-retention cases can already be separated using a relatively simple combination of the inputs (Knauer et al., 2024). Because Random Forest can capture nonlinear correlations and interactions between the simulation parameters while averaging numerous trees to lessen susceptibility to specific cases, it is tried for continuous BMF prediction (Breiman, 2001; Cambioni et al., 2019). Additionally, Gradient Boosting is evaluated since it constructs trees in a sequential manner, with each subsequent tree fixing mistakes produced by the ones before it. This could assist capture more precise changes between various encounter settings (Friedman, 2001; Valencia et al., 2019; Timpe et al., 2020).

To determine whether prediction performance could be improved beyond the baseline models, Gradient Boosting hyperparameters were tuned and physics-derived attributes calculated from the original simulation inputs were tested as a separate feature-engineering experiment. The tuned Gradient Boosting model using the original inputs was retained as the primary model because the derived features produced no meaningful additional gain (Appendix A, Tables A1-A2).

2.3 Model Testing and Performance Metrics

One of the main issues with the archive is that it contains repeated rows from the same physical SPH simulation, for example when the same simulation is post-processed using different FoF linking lengths. Therefore, the model might be trained on one processed version of a simulation and tested on another version of the same simulation using a random row split. 5-fold GroupKFold cross-validation was employed to prevent this. Every out-of-fold (OOF) prediction is made by a model that has not been trained on that physical simulation since all rows from the same physical simulation file are stored in the same fold.

Regression performance is measured using R^2, mean absolute error (MAE), and root mean squared error (RMSE). Classification performance is measured using balanced accuracy, F1, and ROC AUC.

2.4 How the Physical Effects Were Analysed

Although overall scores are helpful, they do not demonstrate whether the model exhibits reasonable physical behaviour in various scenarios. Three complimentary analyses are employed to verify this. To determine how heavily the model depends on each physical input across the dataset, an input-removal test, also known as ablation, first eliminates one original input from the model and analyses how much model performance declines (Section 3.4). Second, to illustrate the direction and form of the model response, controlled tests alter one input while maintaining the others constant (Section 3.5, Figure 3). Third, matched SPH comparisons examine whether an input's significance varies depending on encounter circumstances (Section 3.5, Figure 4).

2.5 When to Trust the Model

The method also determines whether a new setup is comparable to SPH simulations that are already in the archive because model accuracy alone cannot demonstrate the reliability of each individual forecast because an overall score can conceal areas where the model has insufficient data or produces larger errors. When there are multiple independent SPH simulations and a low held-out prediction error in the surrounding area, predictions are more reliable. When the input combination is outside the sampled region, close to the edge of the available data, or poorly sampled, predictions are deemed less accurate. Figure 5 and section 3.6 provide an analysis of these inspections.

The final decision then separates two questions: (1) how strongly the asteroid breaks up, and (2) how much material remains bound to Mars. Breakup is described by the largest-remnant fraction $f_{LR} = M_{largest}/M_{parent}$ where score above 0.90 is mostly intact, 0.50-0.90 is partial breakup, and at or below 0.50 is severe disruption. Retention is measured separately with BMF and Mars-bound mass, $M_{bound} = BMF \times M_{parent}$, this is important because severe breakup can still leave almost no material bound to Mars (Richardson et al., 1998; Walsh & Richardson, 2006).

3. Results

3.1 Exploratory Physical Trends

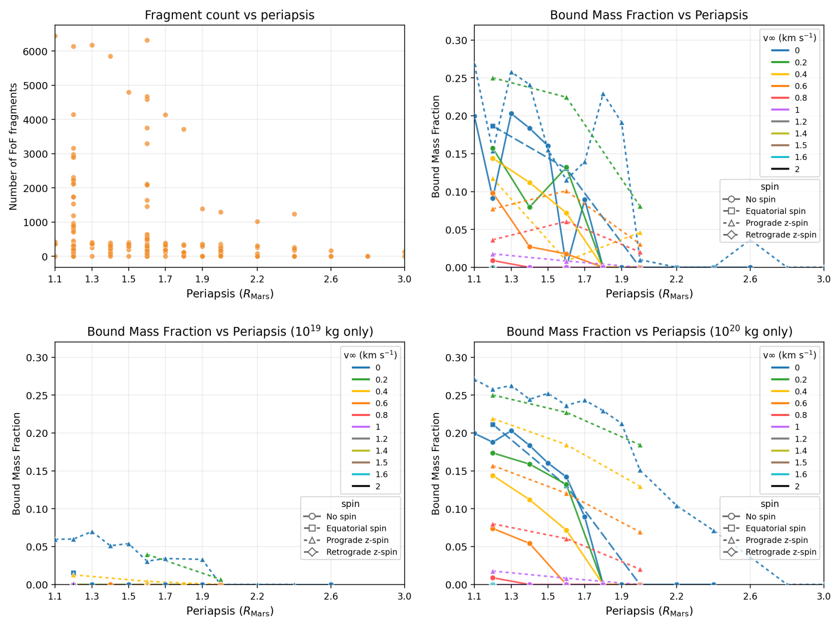

Figure 1. Exploratory SPH trends and mass-dependent bound-retention regimes. (a) Fragment count versus periapsis. (b) BMF across all sampled encounter conditions. (c-d) The same BMF results separated into 10^{19} and 10^{20} kg asteroid-mass groups, showing that the combined scatter includes distinct retention regimes.

Although close interactions do not always result in strong fragmentation or high BMF, Figure 1 illustrates that both fragmentation and bound retention typically decrease as periapsis increases. The mean BMF rises with tiny periapsis and falls with increasing encounter speed. The relative impacts also vary throughout the sampling range: separation between spin states becomes more apparent around 1.7-2.0 Mars radii, whereas velocity differences are more noticeable at roughly 1.1-1.6 Mars radii. The region is too rarely sampled to draw firm conclusions, but one prograde-spin case maintains roughly 3-4% BMF at 2.6 Mars radii whereas the other sampled cases are close to zero.

The combined BMF graph is highly irregular because it has simulations from different asteroid-mass regimes, and separating the mass cases shows that this apparent scatter shows that the bound mass is regime-dependant rather than following a single periapsis trend. The mass-separated panels also show a strong asymmetry in the sampled archive: the 1020 kg cases are much more heavily represented (n=271) than the 1019 kg cases (n=81), and they span substantially higher BMF and remain non-zero over a wider periapsis range. The 10^{19} kg cases are mostly in a low-retention regime. However, because the two mass groups do not contain the same combinations of velocity, spin, and periapsis, this should not be interpreted as a direct causal mass effect. Overall, the results show regime-dependent behaviour, where distinct bound-mass outcomes result from varying relative effects of parameters across the parameter space rather than a single parameter acting independently.

3.2 Eccentricity and Bound Retention

It was previously demonstrated that fragmentation and BMF do not follow a single, straightforward input trend, thus I want to see if relevant physical parameters can more accurately characterize this behaviour. To provide a single physically significant measure of the orbit shape and enable retention to be analysed using the entire encounter trajectory rather than as distinct factors, periapsis and encounter speed are combined to form an encounter eccentricity.

The outcome demonstrates that retention is concentrated close to the parabolic limit, where the eccentricity proxy is almost equal to 1. Any non-zero BMF extends to around 1.127, although strong retention (BMF > 0.10) only manifests up to a proxy of roughly 1.057. As the encounter gets more hyperbolic, the points compress toward BMF = 0, as Figure 2 clearly illustrates. Faster, more hyperbolic fly-bys leave less material bound to Mars because they spend less time in capture-friendly conditions.

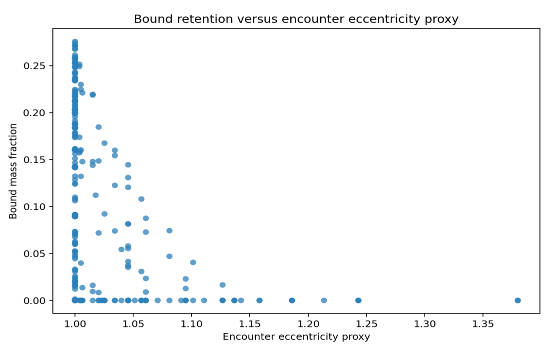

Figure 2. Bound mass fraction versus encounter eccentricity proxy.
Strong and non-zero retention quickly disappears as the encounter becomes more hyperbolic.

3.3 Model Performance and Screening Use

The baseline model comparisons support Gradient Boosting for continuous BMF prediction. Random Forest produces $R^2 = 0.8977$, MAE = 0.0187, and RMSE = 0.0298, while Gradient Boosting achieves $R^2 = 0.9142$, MAE = 0.0175, and RMSE = 0.0272. Gradient Boosting is therefore the stronger baseline model for this dataset.

For the BMF ≥ 10% screening task, Logistic Regression obtains aggregated balanced accuracy of 0.849 and ROC AUC of 0.925, indicating that the two retention groups are reasonably separable. Continuous BMF continues to be the main output; the threshold is simply utilized for prioritisation.

Tuning Gradient Boosting improves performance from $R^2 = 0.9142$, MAE = 0.0175, and RMSE = 0.0272 to $R^2 = 0.9234$, MAE = 0.0159, and RMSE = 0.0257 (Appendix A, Tables A1-A2). Adding the tested physics-derived features produces only a small further change to $R^2 = 0.9240$, MAE = 0.0155, and RMSE = 0.0257. This additional gain is not meaningful, so the tuned Gradient Boosting model using the original simulation inputs is selected as the primary model.

3.4 Overall Physical Parameter Importance

Commented [KN1]: Overall Parameter Importance
Removing periapsis: $\Delta R^2 = -0.574$
Removing velocity: $\Delta R^2 = -0.296$
Removing spin: $\Delta R^2 = -0.316$

Overall Rule
periapsis > spin > velocity

The input-removal test yields the most understandable overall result. The amount of predictive information lost is indicated by the decrease in R^2 when each original physical input is removed from the tuned Gradient Boosting model. Removing periapsis causes the biggest drop ($\Delta R^2 = -0.574$). Removing spin gives $\Delta R^2 = -0.316$, and removing velocity gives $\Delta R^2 = -0.296$. The overall effect of asteroid mass is smaller and less reliable. These values are separate ablation experiments and should not be added together. They display the degree to which the model depends on each input, not whether increasing that input directly raises or lowers BMF. Periapsis is the strongest overall, followed by spin and velocity, although this order may vary locally.

A different response is required for fragmentation. The most extreme fragmentation in the SPH data requires very small periapsis, yet a close encounter does not ensure significant breakdown. FoF connecting length has a direct impact on fragment count as well. This analysis does not provide a single, straightforward global ranking for fragmentation because of this. While the actual fragment pattern depends on the entire encounter and how fragments are gathered during post-processing, periapsis indicates where the largest disruption may occur.

3.5 Checking Physical Trends and Condition-Dependent Effects

The overall ranking of the inputs on which the model is most dependent does not demonstrate how BMF varies with each input or whether an input's significance stays constant under various encounter circumstances. Two follow-up issues are posed in this section: (1) how does BMF change when one input is changed while the others remain unchanged, and (2) do the impacts of significant inputs hold true under various encounter conditions?

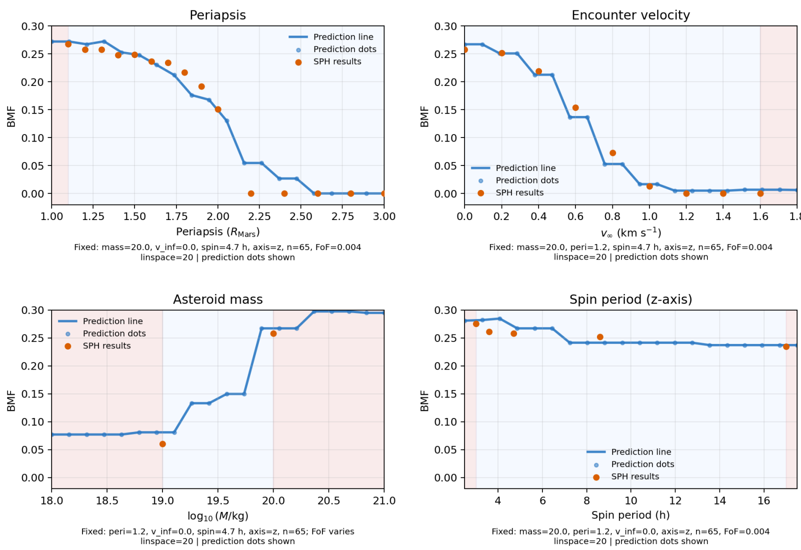

Figure 3. BMF checks where one input changes at a time. Dots are available SPH simulation results and curves show tuned Gradient Boosting predictions with the other inputs fixed. Blue background shows the interpolation range, while red background shows the extrapolation range.

Figure 3 compares the tuned Gradient Boosting response with available SPH simulation results by changing one input at a time while holding the others constant. In the representative periapsis profile, predicted BMF generally decreases from roughly 0.27 at 1.2 Mars radii to approximately zero by about 2.5 Mars radii. The model remains slightly above the SPH results around 2.2-2.4 Mars radii because its prediction is influenced by patterns learned from the wider training data, not periapsis alone. By about 2.5 Mars radii, the input crosses tree thresholds associated mainly with zero-BMF training cases, so the prediction also falls to approximately zero. A similarly clear decline occurs as encounter velocity increases, from roughly 0.25 at low velocity to approximately zero by about 1.2 km/s.

Support for the mass plot is far weaker. There are only roughly two asteroid-mass values with exact SPH scenarios in which every other selected condition is the same. There is insufficient matched data to construct a trustworthy mass trend because the available higher-mass SPH examples are likewise concentrated at low periapsis and low velocity. Therefore, rather of being viewed as a physical threshold, the sharp increase in the Gradient Boosting curve should be considered model dependent. For the z-axis scenario that is shown, the spin-period panel is almost flat, indicating that altering the spin period has minimal impact in this specific configuration. This demonstrates that an input that is significant throughout the whole dataset can only have a minor impact given a particular set of encounter conditions, which does not contradict the overall significance of spin in Section 3.4.

Far outside the sampled parameter ranges, tree-based predictions become stepwise or constant and need not represent physical trends (Appendix Figure A1).

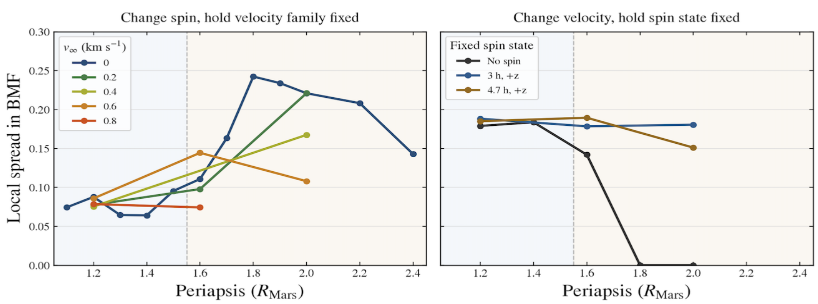

Figure 4. BMF differences within the matched A_{2000}, n_{65} subset at timestep 90000 and FoF linking length 0.004. Left: BMF range across spin states with velocity fixed. Right: BMF range across velocities with spin fixed. These ranges show how strongly outcomes differ within this subset; they do not show direction or prove causation.

Because the archive contains matching situations where these parameters can be changed while mass and other numerical and post-processing settings remain unchanged, spin and velocity are used in Figure 4. Next, periapsis is changed throughout the comparison. When velocity is fixed, the difference between the largest and smallest BMF over all possible spin states is represented by ΔBMF_{spin} in the left panel. In the right panel, spin is fixed and $\Delta BMF_{velocity}$ falls within the same range across all possible velocities. A higher ΔBMF indicates that, in certain circumstances, altering that parameter has a greater impact on the retained mass.

The outcome makes it abundantly evident that each parameter's intensity varies. Changing spin results in a BMF range of around 0.06-0.09 below about 1.5 Mars-radii. This can grow much stronger at bigger periapsis: the spin-related range increases to around 0.24 at 1.8 Mars-radii for v_∞=0, while various other velocity groups reach approximately 0.14-0.22 between 1.6 and 2.0 Mars-radii. The pattern of velocity is distinct. The velocity-related BMF range for the no-spin instances is approximately 0.18 at 1.2-1.4 Mars radii, decreases to approximately 0.14 at 1.6 Mars radii, and reaches zero by 1.8 Mars radii. On the other hand, velocity-related ranges of roughly 0.15-0.19 between 1.6 and 2.0 Mars-radii are still seen in the sampled prograde z-spin situations.

These findings support the importance of regime-dependent parameters: the overall ablation ranking is not universal because spin predominates under some situations while velocity predominates under others.

3.6 Where the Model Can and Cannot Be Trusted

The quantity of local SPH data and prediction error both influence trust. Instead of counting repeated FoF rows, the coverage-error maps count distinct physical simulations and compare that number to the average held-out absolute error in the same area. While sparse or fringe regions can have significantly worse error, regions with many simulations typically have lower error. Low error is insufficient on its own; certain poorly sampled locations appear simple simply because the true BMF is typically zero.

Because the model depends on input combinations, a case may fall inside the range of each individual input and yet have inadequate support. Thus, held-out error and pairwise parameter coverage are compared with local simulation coverage in Figure 5. Individual input ranges conceal unsampled combinations, which are revealed by pairwise mapping. Only when comparable SPH cases are close by in the input space is a prediction regarded as interpolation.

A specific illustration of this issue for the sparsely sampled $10^{19.5}$ kg asteroid mass is provided in Table 2. At periapsis 1.2 and 1.6 Mars radii, when five post-processing rows are provided but only one independent physical SPH simulation supports each periapsis, it compares the true BMF, held-out prediction, and absolute error.

Table 2. Failure case at asteroid mass $10^{19.5}$ kg: the mass is within the sampled range, but only two unique physical SPH simulations support these cases.

Periapsis	**Actual BMF**	**OOF prediction**	**Abs. error**	**Failure**
1.2	0.0913	0.0134	0.0779	Underpredicts
1.6	0.000204	0.0827	0.0825	Overpredicts

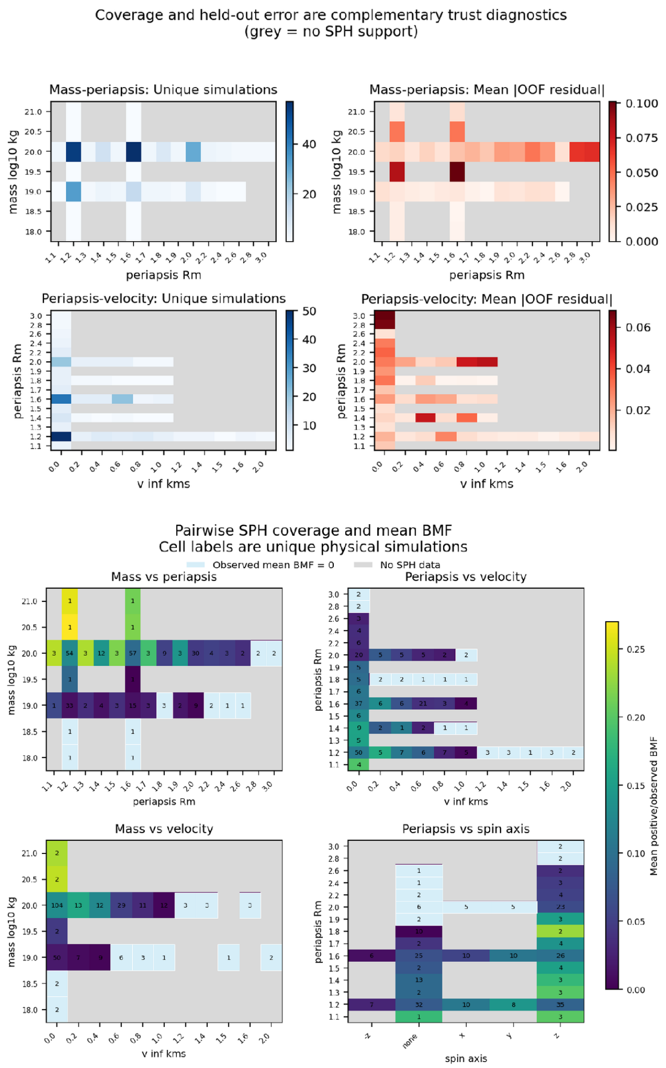

Figure 5. Checks used to judge prediction trust.
(Top: the number of unique SPH simulations and the mean held-out absolute error provide different information. Bottom: pairwise coverage shows input combinations that are missing even when each individual value is within its sampled range.) Grey means no SPH data in that bin, light blue means SPH data exists, but the mean observed BMF is 0, while other coloured cells mean that SPH data exists and mean observed BMF is > 0.

This demonstrates that the held-out model overpredicts the 1.6 Mars-radii scenario by 0.0825 BMF and underpredicts the 1.2 Mars-radii case by 0.0779 BMF. The local relationship has not been successfully learned because the errors are both substantial and have opposite signs. Therefore, a new instance in this region should not be believed just because its mass falls between sampled masses; instead, it should be verified with more SPH.

3.7 How the Model Supports SPH Decisions

So, what is the model actually useful for? The screening process separates disruption from retention. While BMF calculates the amount of material that is still attached to Mars, the largest-remnant fraction indicates whether the asteroid is largely intact, breaks apart somewhat, or is seriously disturbed. Strong disruption can nevertheless result in primarily fleeing debris, which is why both outcomes are distinct.

As a result, the prototype provides the breakup class, estimated Mars-bound mass, expected BMF, data coverage, and a recommendation about the use of SPH. While sparse, edge, transition, or doubtful instances should be examined with SPH, well-supported cases can be swiftly evaluated with ML.

4. Discussion

The clearest periapsis and velocity trends are seen in the controlled experiments, while the matched comparisons show that the effects of spin and velocity vary with encounter conditions. The behaviour in Figure 3 is consistent with the other analyses: although most larger-periapsis cases have near-zero BMF, Figures 1, 4, and 5 show that non-zero retention can still occur under particular combinations of spin, velocity, and mass. This indicates that similar periapsis values can produce different outcomes depending on the full encounter conditions. These relationships are learned from SPH outputs and support interpretation rather than proving physical causality.

The primary physical outcome is that each parameter's impact is contingent upon the other encounter conditions. Within those parameters, periapsis determines the overall strength of the tidal encounter, encounter speed influences the duration of the contact and the favourable capture, and mass and spin alter the outcome. The most obvious periapsis and speed directions are found in the controlled experiments. The paired simulations demonstrate how periapsis and spin state affect the magnitude of the spin and velocity impacts.

Though these associations are acquired from SPH outputs rather than from physical laws, the ML model replicates general trends and reveals interactions throughout the archive. As a result, matched comparisons, controlled tests, and group validation support interpretation but do not prove causality.

As a result, ML's usefulness is restricted. It can swiftly filter well-supported input combinations, identify transition zones, uncover situations that yield promising retained mass, and indicate areas where more simulation could yield valuable information. Long-term debris survivability, precise fragment shapes and distributions from BMF alone, and forecasts for inadequately sampled input combinations cannot all be determined by it. This is evident from the $10^{19.5}$ kg example: when only two independent simulations support the local combination, being numerically inside the training range is insufficient.

Different questions are addressed by the matching and ablation analysis. Although matching cases demonstrate that spin or velocity can become more significant under specific encounter settings, periapsis is generally the strongest.

Compared to BMF, fragmentation requires more prudence. Strong fragmentation is possible but not guaranteed by close periapsis, and the number of pieces is directly altered by the FoF connecting length. Therefore, rather than taking the place of BMF, fragment count and largest-remnant metrics offer additional information. Strong breakup does not always result in much bound debris, therefore the procedure separates breakup severity from Mars-bound retention.

All things considered, the ML model is a quick screening method for the current SPH archive. Poorly sampled cases, transition regions, and issues requiring certain fragment attributes still require SPH.

5. Conclusions, Limitations and Future Work

5.1 Conclusions and Answers to the Research Questions

This study sought to determine which inputs are most important for fragment creation and retained debris, how orbital and physical characteristics affect disruption and bound retention, and whether machine learning (ML) can forecast outcomes while indicating when its predictions should be believed. Below is a summary of the responses.

Question 1: How do orbital and physical parameters affect disruption and bound retention? Although the impact varies depending on the other inputs, closer and slower contacts typically retain more debris. In the representative controlled test, predicted BMF generally decreases from roughly 27% at 1.2 Mars radii to approximately zero by about 2.5 Mars radii as periapsis increases. A similarly clear decline occurs as encounter speed increases, from roughly 25% at low velocity to approximately zero by about 1.2 km/s. Rather than being universal thresholds, these reactions are condition specific. Mass and spin alter the outcome, speed alters the encounter and capture conditions, and periapsis regulates tidal power.

Question 2: Which inputs matter most for retained debris? There is no single universal ranking under all encounter conditions. For BMF, the tuned Gradient Boosting input-removal analysis identifies periapsis as the strongest overall physical input, with its removal causing the largest decrease in performance ($\Delta R^2 = -0.574$), followed by spin (-0.316) and velocity (-0.296). However, matched cases show that their relative importance changes locally.

Question 3: Can machine-learning models predict outcomes within the sampled data while showing when a prediction should or should not be trusted? Yes, in cases where the SPH archive supports comparable input combinations. The primary tuned Gradient Boosting model using the original simulation inputs obtains a grouped cross-validated R^2 of 0.9234, MAE of 0.0159, and RMSE of 0.0257. Adding physics-derived features produces only a small further change to $R^2 = 0.9240$ and MAE = 0.0155, with RMSE unchanged at 0.0257, so the simpler tuned model is retained. Performance is not equally dependable everywhere, however. Despite the input values falling within the numerical training ranges, held-out errors increase to roughly 0.08 BMF, or 8 percentage points, in the sparsely sampled $10^{19.5}$ kg example. This demonstrates that being within the overall input range is insufficient for a reliable prediction; comparable SPH cases must also exist nearby in the sampled parameter space. Thus, ML can support rapid screening and simulation prioritisation, while sparse, edge, transition, or poorly supported cases still require additional SPH investigation.

Overall, the project shows that tidal-disruption outcomes depend on interacting physical conditions rather than one universal ranking of parameters. Machine learning makes the existing SPH archive faster to explore, shows where predictions are well supported, and highlights where new simulations would be most useful. It complements SPH rather than replacing the simulation physics.

5.2 Limitations

The simulation archive's short size and uneven sampling are its primary drawbacks. A single SPH simulation can be analysed in multiple ways, hence many rows are not independent physical evidence. Although it is unable to produce missing simulations, group validation keeps the test score from becoming unduly optimistic. Even when each individual input value falls between its reported minimum and maximum, predictions are still unclear for combinations that are poorly sampled, close to the data's edges, or outside the combinations observed during training.

Additionally, instead of solving the physical laws directly, the ML model learns patterns from SPH results. It does not follow the subsequent orbits of fragments and does not impose conservation of mass, momentum, energy, or angular momentum. Therefore, a reliable BMF prediction cannot determine whether debris accumulates into moons, forms a disk, or endures for a long time. Additionally, fragment measurements are more sensitive to the FoF connecting length employed in post-processing and are noisier than BMF.

Higher grouped-validation scores are obtained by exploratory XGBoost and hurdle models, but more testing is necessary before they can take the place of the primary Gradient Boosting. Because outcome-derived quantities are not known prior to simulation, models that employ them are not appropriate for pre-SPH screening.

5.3 Future Work

The highest priority for future work is to run new SPH simulations in poorly covered mass-periapsis-velocity-spin combinations, especially near the change between retained and zero BMF. More matched groups should repeat the A_{2000}, n_{65} comparison at other masses and resolutions while keeping processing choices fixed. It would be possible to determine whether the stronger spin disparities also occur in other groups by reporting the BMF range together with the actual BMF values and the number of independent simulations. It would be more robust to test the model on whole new SPH runs created after training rather than repeatedly dividing the current library.

Future research should enhance automatic identification of input combinations with inadequate support and per-prediction reliability estimates. Before updating the current model, new SPH simulations should be used to test exploratory XGBoost and hurdle models. Stronger physical limitations, multi-output prediction, and uncertainty estimates are possible additional enhancements.

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Appendix A. Supporting Model Results

Table A1. Model tuning parameters and physics-derived features investigated.

Category	Parameters	Baseline raw input(s)	Changed / calculation	Effect on performance
Model tuning	Number of trees (n_estimators)	200	500	Increased R^2 by 0.0092
	Learning rate (learning_rate)	0.1	0.08	
	Subsample	1	0.8	
	Minimum samples per leaf (min_sample_leaf)	2	1	
Simple derived feature	Encounter speed squared	v_∞	v_∞^2	Increased R^2 by 0.001 (spin frequency) and 0.002 (inverse periapsis)
	Inverse periapsis	r_p	$1/r_p$	
	Spin frequency	P_{spin}	$1/P_{spin}$	
	Asteroid radius	$M_{asteroid}$	$R = \left(\frac{3M}{4\pi\rho_{ast}}\right)^{1/3}$ with ρ_{ast} = 2700 kg/m^3	
Combined physics feature	Periapsis-speed interaction	r_p , v_∞	$r_p \times v_\infty$	Increased R^2 by 0.002
	Encounter eccentricity	r_p , v_∞	$e = 1 + \frac{r_p v_\infty^2}{\mu_{Mars}}$	
	Time within 2 R_{Mars}	r_p , v_∞	$t_{inside}(r_p , v_\infty; 2R_{Mars})$	
	Time within tidal-disruption radius	r_p , v_∞	$t_{inside}(r_p , v_\infty; R_{tidal})$	

Table A2. Comparison of BMF regression models using five-fold GroupKFold by physical SPH simulation, including exploratory advanced models to show the potential for further performance improvement in future work.

Model	**R^2**	**MAE**	**RMSE**	**Status**
Baseline Random Forest (RF)	0.8977	0.0187	0.0298	Baseline model
Baseline Gradient Boosting (GB)	0.9142	0.0175	0.0272	Best baseline model
Tuned GB	0.9234	0.0159	0.0257	Main model
Tuned GB + derived features	0.9240	0.0155	0.0257	No meaningful gain
XGBoost	0.9377	0.0132	0.0232	(Exploratory only) strongest single-stage alternative
Two-stage CatBoost hurdle	0.9483	0.0122	0.0212	(Exploratory only) lowest MAE among hurdle tests
Two-stage NGBoost hurdle	0.9485	0.0127	0.0211	(Exploratory only) highest R^2 / lowest RMSE

The 10% BMF threshold (using logistic regression) is used for screening rather than as a criterion for moon formation, therefore excluded from the table. More sophisticated alternatives, including XGBoost and two-stage hurdle models, were also investigated but were retained as exploratory models, however they remained exploratory and are not covered in more detail in this project as they introduce additional model complexity, particularly the two-stage approaches, and would require further testing and interpretation before replacing the primary model. This was beyond the scope of the present project since the primary purpose was to demonstrate the possibility of further model improvement in subsequent work rather than to serve as the project's primary outcome.

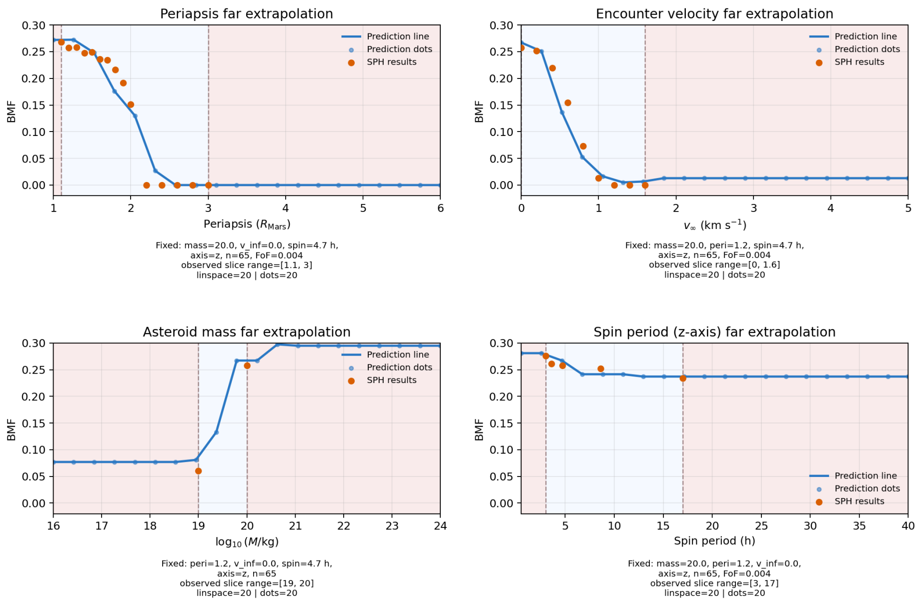

Figure A1. Gradient Boosting predictions beyond the observed SPH parameter ranges.

The shaded central regions indicate the ranges represented by the corresponding SPH slice. Outside these ranges, the tree-based model does not extrapolate a physical trend. Instead, it continues to use partitions learned from the training data, producing stepwise or constant predictions, as seen most clearly for periapsis, velocity, and spin. For mass, the prediction initially increases slightly beyond the observed range as the changing mass and associated derived features continue to cross learned tree thresholds, before eventually remaining within the same terminal partitions and settling onto a near-constant plateau. This does not occur for the other parameter sweeps because their out-of-range values enter terminal partitions more immediately, after which further changes no longer alter the prediction. These out-of-range predictions should therefore not be interpreted as physical trends.

Appendix B. Decision-Support Prototype

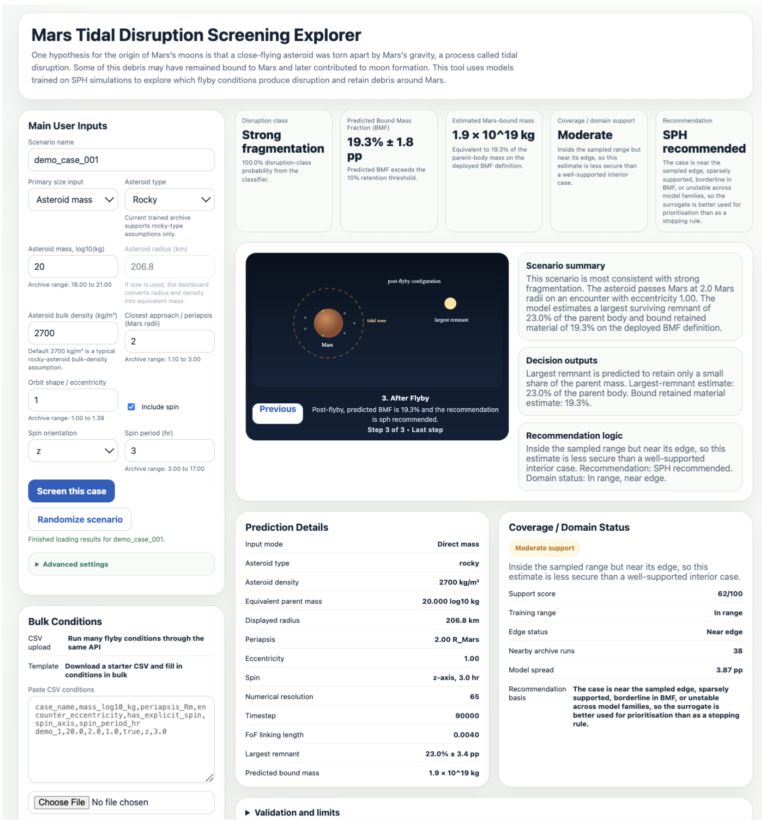

Figure B1. Mars Tidal Disruption Screening Explorer. The demonstration interface uses the main Random Forest model to report predicted BMF, estimated Mars-bound mass, data-coverage checks and a recommendation on whether another SPH simulation is needed.

A demonstration implementation of the screening workflow was developed as a dashboard and API package. It packages the model, input processing, coverage checks and screening rules into a runnable interface. The application can be downloaded and run locally by cloning the project repository.

Appendix C. AI acknowledgement Statement

I used OpenAI Codex (https://chatgpt.com/codex) throughout the project for targeted code-development support. This included producing initial implementation drafts for data extraction and processing, exploratory data analysis, feature engineering, diagnostic and visualisation scripts, refactoring, debugging, repository documentation, and some commit messages. All AI-assisted code and documentation were reviewed, modified, tested, and verified by me before inclusion. I specified the scientific questions, modelling targets, physics-derived features, thresholds, evaluation strategy, plot selection, interpretation, and final project decisions, and I can explain, modify, and reproduce the submitted implementation. AI was not used to write the Project Plan or Final Report. The scientific conclusions, design decisions, and submitted report are made by me and reflect my own work and understanding. Further details of AI usage are documented in docs/ai_usage.md.